

nereus_Pi_shield_ADIN2111 — BOM for review

PCB-000001-AA rev AA · from docs/design-review/bom.csv · 2026-10-07 · 51 lines, 105 fitted parts per board (+ 1 DNP) · 11 lines where the planned part differs from Sofar's (for Sofar's review; nothing adopted before it, D26/D27)

Refs	Qty	Sofar's part (specified)	Planned part	Why (planned note)	Backups (Alt1 / Alt2)	Sourcing	Footprint
C1, C3, C9-C15, C28, C53	11	CL05A104KA5NNNC Samsung	CL05A104KA5NNNC Samsung LCSC C100072	As specified; in LCSC stock (checked 2026-10-06)			FP-CL278-IPC_C
C2, C4	2	CL05A105KP5NNNC Samsung	CL05A105KP5NNNC Samsung LCSC C14445	As specified; in LCSC stock (checked 2026-10-06)			FP-CL511-IPC_C
C5, C6	2	CC0402KRX5R6BB684 YAGEO	CC0402KRX5R6BB684 Yageo LCSC C113784	As specified; in LCSC stock (checked 2026-10-06)			FP-CC0402-MFG
C7, C8	2	CC0402JRNPO9BN150 Yageo	CC0402JRNPO9BN150 Yageo LCSC C106997	As specified; in LCSC stock (checked 2026-10-06)			FP-CC0402-0_55-IPC_B
C16, C18, C24, C25	4	08051C474KAT2A KYOCERA AVX	CC0805KKX7R0BB474 YAGEO LCSC C596323	Same spec, 470 nF ±10 % 100 V X7R 0805, largest LCSC stock (250,730 vs the AVX ALT2's 750, 2026-10-06); specified AVX has no LCSC listing; PoDL front end: Sofar review	CC0805KKX7R0BB474 (YAGEO, C596323) 08051C474K4Z2A (KYOCERA AVX, C597303) <i>470 nF ±10 % 100 V X7R 0805; PoDL front end: Sofar review</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	FP-0805-L_2_01_0_2-W_1_25-MFG
C17, C26	2	C2012X7S2A105K125AB TDK	C2012X7S2A105KT000N TDK LCSC C342785	TDK C2012 series, same 1 µF ±10 % 100 V X7S 0805 per LCSC (thickness not confirmed: K125AB is 1.25 mm, KT000N doesn't state it); specified number out of stock at LCSC; power path: Sofar review	C2012X7S2A105KT000N (TDK, C342785) GCM21BC72A105KE36L (Murata, C126585) <i>1 µF ±10 % 100 V X7S 0805; power path: Sofar review</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	FP-C2012-125-0_2-MFG
C19, C27	2	C0805C102MDRACTU Kemet	C0805C102KDRACTU KEMET LCSC C2167549	Same KEMET series, ±10 % instead of ±20 % (tighter); 1 nF 1 kV X7R 0805; specified ±20 % part has no LCSC listing; PoDL front end: Sofar review	C0805C102KDRACTU (KEMET, C2167549) 0805B102K102CT (Walsin, C303890) <i>1 nF 1 kV X7R 0805, ±10 % (tighter); PoDL front end: Sofar review</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	CAPC2013X145X50LL20T25
C20, C29, C49, C50, C54	5	GRM188R72A104KA35D Murata	GRM188R72A104KA35D Murata LCSC C77058	As specified; in LCSC stock (checked 2026-10-06)			FP-GRM188-0_1-e0_2_0_5-IPC_B
C21	1	CL32B106KBJNNNE Samsung	UMK325AB7106KM-T Taiyo Yuden LCSC C386167	10 µF ±10 % 50 V X7R 1210 from Taiyo Yuden, an established MLCC maker, for the power path (FH ALT1 has more stock, kept as backup); specified Samsung out of stock at LCSC and DK; Sofar review	1210B106K500NT (FH, C116808) UMK325AB7106KM-T (Taiyo Yuden, C386167) <i>10 µF ±10 % 50 V X7R 1210; power path: Sofar review</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	FP-CL32-IPC_C
C22, C23	2	EEEFT1H470AP Panasonic	EEEFT1H470AP Panasonic LCSC C92059	As specified (Sofar review): no potting-safe like-for-like alternate (constraint 8); LCSC listing out of stock: JLC global sourcing		JLC global sourcing (first build) until an ALT is approved (D26)	FP-FTD-MFG
C30, C55	2	UMK107BBJ225KA-T Taiyo Yuden	UMK107BBJ225KA-T Taiyo Yuden LCSC C268016	As specified; in LCSC stock (checked 2026-10-06); Taiyo Yuden now lists it as MSASU168BB5225KTNA01			FP-0603-L_1_6_0_2-W_0_8_0-MFG
C31	1	C1608X5R0J226M080AC TDK	CL10A226MQ8NRNC Samsung LCSC C59461	Same spec, 22 µF ±20 % 6.3 V X5R 0603; specified TDK out of stock at LCSC (and DK until 2027-01)	CL10A226MQ8NRNC (Samsung, C59461) GRM188R60J226MEA0D (Murata, C77042) <i>22 µF ±20 % 6.3 V X5R 0603</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	CAPC1608X100X20ML10
C32	1	GRJ155R60J106ME11D Murata	GRJ155R60J106ME11D Murata LCSC C469527	As specified; in LCSC stock (checked 2026-10-06)			FP-GRJ155-0_2-MFG

C33	1	GRM155R60J475ME87D Murata	GRM155R60J475ME87D Murata LCSC C76996	As specified; in LCSC stock (checked 2026-10-06)			FP-GRM155-0_15-MFG
C56, C57	2	GRM21BR61C226ME44L Murata	GRM21BR61C226ME44L Murata LCSC C86817	As specified; in LCSC stock (checked 2026-10-06)			C_0805_2012Metric
C58	1	GRM155R61A475MEAAD Murata	GRM155R61A475MEAAD Murata LCSC C335105	As specified; in LCSC stock (checked 2026-10-06)			C_0402_1005Metric
D1	1	SMA6F33A STMicroelectronics	SMA6F33A-M3/H Vishay	Sofar's own listed second source (their MANUFACTURERPARTNUMBER1); ST SMA6F33A is obsolete (DK); no LCSC listing found (2026-10-06): JLC global sourcing; Sofar Q7		JLC global sourcing of Sofar's second source Vishay SMA6F33A-M3/H (ST part obsolete; Sofar Q7) (D26)	SMA6F33A-Footprint-1
D2, D3	2	RB058LAM-60TFTR ROHM	RB058LAM-60TFTR ROHM	As specified; JLC global sourcing (D26): no LCSC listing		JLC global sourcing (D26); no LCSC listing	FP-PMDTM-MFG
D4, D5	2	CDSOD323-T36SC Bourns Inc.	CDSOD323-T36SC Bourns LCSC C911382	As specified; in LCSC stock (checked 2026-10-06)			FP-SOD-323-MFG
D8	1	KT-0603R Hubei KENTO Elec	KT-0603R Hubei KENTO LCSC C2286	As specified; in LCSC stock (checked 2026-10-06)			LED_0603_1608Metric
D9, D10	2	KT-0603YG Hubei KENTO Elec	KT-0603YG Hubei KENTO LCSC C2289	As specified; in LCSC stock (checked 2026-10-06)			LED_0603_1608Metric
J1	1	Raspberry_Pi_2_3	TBD	Part not chosen: 2x20 2.54 mm female TH socket, height open (Nick)		Part not chosen: 2x20 2.54 mm female TH socket, height open (Nick). Candidates ZHOURI 2.54-2*20 C2977589, Megastar ZX-PM2.54-2-20PY C7499354 (both 8.5 mm)	PinSocket_2x20_P2.54mm_Vertical
J5	1	SM02B-GHS-TB JST	SM02B-GHS-TB JST LCSC C189893	As specified; LCSC listing out of stock (2026-10-06): JLC global sourcing		JLC global sourcing (D26); LCSC C189893 out of stock, DK 10,839	JST_GH_SM02B-GHS-TB_1x02-1MP_P1.25mm_Horizontal
L1, L2	2	SRF1260-101M Bourns	SRF1260-101M Bourns LCSC C6496096	As specified; in LCSC stock (checked 2026-10-06) (thin margin)			SRF1260-101M-Footprint-1
L3, L6	2	PA5432.822NLT Pulse (YAGEO group)	PA5432.822NLT Pulse (YAGEO group)	As specified; JLC global sourcing (D26): no LCSC listing		JLC global sourcing (D26); no LCSC listing	FP-PA5432_472NLT-MFG
L4	1	DFE201612E-2R2M=P2 Murata	DFE201612E-2R2M=P2 Murata LCSC C337893	As specified; in LCSC stock (checked 2026-10-06)			FP-DFE201612C-MFG
MP1-MP4	4	78614015360 Würth Elektronik	78614015360 Würth Elektronik	As specified; JLC global sourcing (D26): no LCSC listing		JLC global sourcing (D26); no LCSC listing, DK 7,835	78614015360-Footprint-2
R1, R44-R46	4	RC0402FR-071K5L Yageo / Phycomp	RC0402FR-071K5L Yageo LCSC C114759	As specified; in LCSC stock (checked 2026-10-06)			RESC1005X40X25ML05T10
R2-R6, R26, R27	7	ERJ-2RK4701X Panasonic	ERJ-2RK4701X Panasonic LCSC C400631	As specified; in LCSC stock (checked 2026-10-06)			RESC1005X40X25LL05T05
R7	1	RC0402JR-0710KL Yageo	RC0402JR-0710KL Yageo LCSC C60489	As specified; in LCSC stock (checked 2026-10-06)			FP-RC0402-0_4-IPC_C

R8	1	UR73D1JT TD10L0F KOA Speer	UR73D1JT TD10L0F KOA Speer	As specified (Sofar review), JLC global sourcing: no LCSC listing. ROHM PMR03 (ALT1) electrodes (b 0.35 mm, datasheet Rev.PMR03-IA-012E) sit inside these pads, which were drawn for the UR73D1J's 0.55 mm electrodes and reach 0.35 mm under the PMR03 body; ROHM gives no land pattern, so not adopted	PMR03EZPFU10L0 (ROHM, C308571) WSLP0603R0100FEA (Vishay, C5334451) <i>Sofar review: power-path current sense; check land pattern</i>	JLC global sourcing (first build) until an ALT is approved (D26)	RESC1608X60X55ML20T10
R11 (DNP)	1	CRCW12100000Z0EA Vishay Dale	CRCW12100000Z0EA Vishay Dale LCSC C844880	As specified; DNP (not fitted)			FP-CRCW1210-e3-IPC_B
R12, R13, R17, R19	4	CRCW12100000Z0EA Vishay Dale	CRCW12100000Z0EA Vishay Dale LCSC C844880	As specified; in LCSC stock (checked 2026-10-06)			FP-CRCW1210-e3-IPC_B
R14, R18	2	RR0510P-101-D Susumu	RT0402BRD07100RL YAGEO LCSC C705627	100 Ω thin film 0402, ± 0.1 % ± 25 ppm (tighter than the specified ± 0.5 %); specified Susumu has no LCSC listing; PoDL front end: Sofar review	RT0402BRD07100RL (YAGEO, C705627) ERA2AEB101X (Panasonic, C445575) <i>100 Ω ± 0.1 % ± 25 ppm thin film 0402 (tighter); ALT2 AEC-Q200; PoDL front end: Sofar review</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	FP-RR0510-IPC_B
R15, R16	2	CRCW08057R50FKEAHP Vishay	CRCW08057R50FKEAHP Vishay Dale	As specified (Sofar review): the only stocked alternate (YAGEO SR0805, ALT1) has no stated pulse rating and ± 200 vs ± 100 ppm/ $^{\circ}$ C; JLC global sourcing: no LCSC listing	SR0805FR-477R5L (YAGEO, C873769) <i>Sofar review: 7.5 Ω ± 1 % 0.5 W, ± 200 vs ± 100 ppm/$^{\circ}$C, pulse rating not stated</i>	JLC global sourcing (first build) until an ALT is approved (D26)	CRCW08057R50FKEAHP-Footprint-1
R20, R39	2	CRCW04021M82FKED Vishay Dale	RTT021824FTH RALEC LCSC C166520	Only stocked LCSC match: 1.82 M Ω ± 1 % 62.5 mW 50 V ± 100 ppm 0402; specified Vishay part has no LCSC listing	RTT021824FTH (RALEC, C166520) <i>1.82 MΩ ± 1 % 62.5 mW 50 V ± 100 ppm</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	FP-CRCW0402-e3-MFG
R21, R35, R37, R43	4	RMCF0402FT100K Stackpole	0402WGF1003TCE UNI-ROYAL LCSC C25741	JLC Basic part (no extended-part fee), ample LCSC stock; 100 k Ω ± 1 % 62.5 mW 50 V ± 100 ppm 0402; specified Stackpole is out of stock at LCSC and is AEC-Q200, this one doesn't state it	0402WGF1003TCE (UNI-ROYAL, C25741) RC0402FR-07100KL (YAGEO, C60491) <i>100 kΩ ± 1 % 62.5 mW 50 V ± 100 ppm; ALT1 is JLC Basic; AEC-Q200 not stated (Stackpole is)</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	FP-RMCF0402-IPC_C
R22, R40	2	RC0402FR-07200KL Yageo / Phycomp	RC0402FR-07200KL Yageo LCSC C114763	As specified; in LCSC stock (checked 2026-10-06)			RESC1005X40X25LL05T10
R23	1	RC0402FR-0722K1L Yageo	RC0402FR-0722K1L Yageo LCSC C163478	As specified; in LCSC stock (checked 2026-10-06)			FP-RC0402-0_4-IPC_B
R34	1	RC0402FR-079K09L Yageo / Phycomp	AC0402FR-079K09L YAGEO LCSC C227284	Same YAGEO 0402 thick-film family, 9.09 k Ω ± 1 % 62.5 mW 50 V ± 100 ppm; specified part has no LCSC listing; our part (D18), no amendment needed	AC0402FR-079K09L (YAGEO, C227284) 0402WGF9091TCE (UNI-ROYAL, C11549) <i>9.09 kΩ ± 1 % 62.5 mW 50 V ± 100 ppm thick film; our part (D18), no amendment</i>	Planned part from LCSC once Sofar approves it (D27); until then the specified part via JLC global sourcing	RESC1005X40X25LL05T10
R38	1	RC0402FR-0713K7L Yageo	RC0402FR-0713K7L Yageo LCSC C274888	As specified; in LCSC stock (checked 2026-10-06)			FP-RC0402-0_4-IPC_B
R41	1	RC0402FR-071ML Yageo	RC0402FR-071ML Yageo LCSC C138033	As specified; in LCSC stock (checked 2026-10-06)			R_0402_1005Metric
R42	1	RMCF0402FT10K0 Stackpole	RMCF0402FT10K0 Stackpole LCSC C6111655	As specified; in LCSC stock (checked 2026-10-06)			FP-RMCF0402-IPC_C
T1, T2	2	74930000 Würth Elektronik	74930000 Würth Elektronik LCSC C4354923	As specified; in LCSC stock (checked 2026-10-06)			FP-74930000-MFG

U1	1	ADIN2111BCPZ Analog Devices	ADIN2111BCPZ Analog Devices	As specified; JLC global sourcing (D26): no LCSC listing		JLC global sourcing (D26); no LCSC listing, DK 120	ADIN2111BCPZ-Footprint-1
U2, U3	2	AP22913CN4-7 Diodes Inc.	AP22913CN4-7 Diodes Inc. LCSC C2150198	As specified; LCSC has 30 of the 40 needed (2026-10-06): JLC global sourcing for the rest		JLC global sourcing, all 40 (LCSC has 30) (D26)	AP22913CN4-7-Footprint-1
U4	1	INA232AIDDFR Texas Instruments	INA232AIDDFR Texas Instruments LCSC C5447660	As specified; in LCSC stock (checked 2026-10-06)			INA232AIDDFR-Footprint-1
U5, U10	2	LMR51430YDDCR Texas Instruments	LMR51430YDDCR Texas Instruments LCSC C5210749	As specified; in LCSC stock (checked 2026-10-06)			SOT-23-THIN-6_DDC_TEX
U6	1	TPS62840YBGR Texas Instruments	TPS62840YBGR Texas Instruments LCSC C2071139	As specified; in LCSC stock (checked 2026-10-06)			FP-YBG0006-MFG
U11	1	TPS26621DRCR Texas Instruments	TPS26621DRCR Texas Instruments LCSC C1848341	As specified; in LCSC stock (checked 2026-10-06)			Texas_DRC0010J
Y1	1	ABM11W-25.0000MHZ-6-K1Z-T3 Abracon	ABM11W-25.0000MHZ-6-K1Z-T3 Abracon	As specified; JLC global sourcing (D26): no LCSC listing		JLC global sourcing (D26); no LCSC listing, DK 87	FP-ABM11W-MFG